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US Corporate Bond Market Liquidity: Should insurers be concerned?

Is corporate

bond liquidity a potential problem for the market? Most investment professionals define "Liquidity" as a market's ability to transact buys and sells at a reasonable price, within a reasonable time. Liquidity is often measured by looking at the spread between

bond bid and ask prices, or spread. The bid-ask spread is the difference between the price at which broker/dealers are willing to buy (bid) and the price at which broker/dealers are willing to sell (ask). Spread widening would indicate that there is less liquidity in the market. An alternative measure is the

bond market's turnover rate. This rate is calculated using the ratio of trading volume to outstanding assets. A lower turnover rate means less liquidity.

Trading Bonds: The Role of Brokers and Dealers

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Economic Review

The Bond Market

In contrast

to the second quarter of this year, the Treasury yield curve was little changed on the short end (1-3 years), while intermediate and longer dated Treasury yields decreased significantly. At the end of June, the 10 year and 30 year US Treasury notes yielded 2.35% and 3.11% respectively. These same securities were yielding 2.06% and 2.87% by the end of September. As a result, relative performance favored bonds of longer maturity and higher quality. The best performing sector of the Barclays Aggregate Index was the US Treasury: 20+ year with a return of 5.32%. The worst performing sector was the long foreign agency with a return of -8.75%. Clearly, sector performance favored the safety

of US sovereign bonds to those issued by foreign sovereigns and their agencies. Credit quality returns also favored bonds of higher quality. The lowest rung of investment grade bonds rated BBB had negative performance in the third quarter (-0.73%). The US Corporate High Yield Index published by Barclays showed returns of -4.86% for the quarter and -2.45% year-to-date, as of September 30.

What Happened to the Dual Mandate?

The Federal Reserve's decision to keep the fed funds overnight rate in the 0-0.25% range on September 17 was a significant event. That decision seems to support the notion that the Committee is no longer exclusively focused on the dual mandate of full employment and stable prices (inflation).

Perhaps we should add "market stability" to the Fed's agenda? From the post-meeting statement we focus on the following segment:

"Consistent with its statutory mandate, the Committee seeks to foster maximum employment and price stability. Recent global economic and financial developments may restrain economic activity somewhat and are likely to put further downward pressure on inflation in the near term."

The decision to keep the overnight rate unchanged was a surprise to the majority of economists and market participants. The Federal Reserve's board members have been preparing the market for an increase in



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“The decision to keep the overnight rate unchanged was a surprise to the majority of economists and market participants.”

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short-term rates for most of the year. Steady improvement in unemployment and benign inflation pressures have been consistent with the message from the Fed that there is no longer a need for emergency monetary policy interest rates to remain at zero. Is the Fed losing credibility with the market? Perhaps that may explain why capital markets of both stocks and bonds reacted negatively to the news that day. One of the main reasons that our central bank is the leader in effecting monetary policy worldwide is because of its credibility and discipline to economic fundamentals. It is perceived to be politically unbiased and true to its mandates. So, its decision to keep monetary policy easy when it has been transmitting a tighter stance was met with disappointment. The capital markets are wondering whether its agenda has changed to encompass a greater sensitivity to global financial stability than to its own economy, and increased volatility in both the stock and bond markets has been the result (as the sector performance discussed above has indicated).

There are several challenges to global growth, and certainly too many to effectively address in great detail here. But we do need to mention the key drivers of recent market volatility and why we think they will continue to play a role in future market expectations.

China's slowing growth

There is much discussion in the financial media regarding China's prospects for growth and its financial stability. While it is true that measures of growth have diminished, they are still at higher rates than all of the developed economies. The current GDP growth rate is measuring 7% annually. However, that is down from 7.4% in 2014 and 7.7% in 2013.

Collapsing commodity prices

As mentioned above, the slowing growth in China has translated into less demand and lower commodity prices. The biggest declines have come in the metals and energy sectors. Industrial metals, such as copper, aluminum, and iron ore, are down significantly.

Declining oil prices

The days of \$90–\$100 per barrel of oil are gone for the foreseeable future. This is a positive development for net importers of oil, but quite a problem for those net exporters.

Greece's debt problem

The fundamental issues have not been resolved; they've just been postponed. The deferral of implementing a longer-term solution to the Greek debt problem will continue to stoke the embers of uncertainty for the fate of the

euro zone and its currency.

It is easier to understand why the Federal Reserve was hesitant to move off its easy monetary policy when we consider the above factors, especially combined with a US labor market that may be weaker than data suggests. With unemployment at 5.1% and approaching what historically would be considered the natural rate of unemployment, why has our economy failed to produce higher growth rates? The health of our labor market can no longer be defined by the unemployment rate alone. We need to acknowledge that we still have a sizable number of people working part-time who wish to be full-time. The underemployment rate is still 10.3%. We also have the lowest labor participation rate since 1977. Wage growth is less than spectacular at 2.2% year-over-year. The average number of weekly hours worked is pretty stagnant at 34.6. So, while our economy has added jobs at a decent pace since the recession ended, these are not high paying jobs, and wage inflation is benign.

As long as these uncertainties persist, volatility will remain high and interest rates low. The Federal Reserve may have assumed an implicit mandate of being the stabilizing influence in the global markets.

Economic Review

Unfortunately, it is the savers who will continue to be disadvantaged by these policies.

The Equity Market

The third quarter closed after more than a month of downtrodden markets and moments of extreme volatility. There have been several catalysts at play, but namely China and actions by our own central bank have played a large role in people's worries and fears as of late.

China

The downturn started over concerns about China's economic growth slowing down and dragging the world into a recession. This is something that has been brewing for a while, but was pushed into high gear after the country's stock market crash in late August. Because China is not the most transparent country in the world, there is the additional fear that economic indicators may not have been accurately stated by the Chinese government for some time. After the crash, in an effort to make exports more competitive, the government devalued its currency, further shocking the world and causing fears about a possible currency war, and raising more concern about how bad the situation really is in China. As has become the norm in recent years, this kind of news tends to send shock waves through the markets while investors sort out a likely

outcome.

The Fed's Power to Sway

We believe one can conclude that global economic stability is clearly very important to the Fed, perhaps more than anyone thought. And it seems true that investors perceive the global economy to be weaker than what was previously priced into the markets. From a global monetary policy perspective, the Fed's decision may inspire other central banks to protect against further degradation in China.

The World in Numbers

The third quarter was nothing short of dismal for most of the globe. In the US, the S&P 500 Index was down 6% over the quarter, and was at one time down over 1000 points intraday, and that was actually better than the rest of the world combined, and particularly the emerging markets, which fell by a whopping 18.5%. Even commodities took a beating this quarter, including, unsurprisingly, oil.

A Talk About Corrections

A correction is defined as a drop in an index of 10% from its peak, a scenario we experienced this quarter. Broadly speaking, corrections are small and short lived; they are blips in an otherwise strong economy. Historically, corrections have tended to occur

every 18 months or so, and the last one we experienced was in 2011. In this way, we were long overdue for a correction, which is why it's not shocking that it occurred, although we realize that doesn't make the blow any easier. When we look at the history of the stock market performance, most of the time, the big, protracted drops have occurred either during a recession or at the end of a tightening cycle. Given the current zero interest rate environment, the more relevant question seems to be whether we are approaching a recession. In looking at the consensus US economic forecast, a recession is not anticipated in the foreseeable future. Instead, a slow growth, low inflation environment is expected to persist. Although this economic picture and diagnosis could change, we believe that the indicators point to the markets enduring a correction that, if a history of corrections is any indication, will be made up for in the upcoming months, rather than a crash.

Outlook

As bond yields are expected to remain low for the near future, we continue to favor stocks over bonds, and domestic stocks in particular longer-term. We believe that while the US economy will continue to grow, a

slowdown in China and decreasing energy prices will be headwinds for many foreign economies, specifically those who are significant commodity producers or those with significant exposure to China. However, there is no doubt that the current volatility and concern being felt will continue to be headwinds for stocks for some time. We will use the cash extracted from a few strategic sells towards names we feel better about in this environment going forward. Although it is difficult to forecast where the stock market will bottom, we believe this is more of a buying opportunity than a sustained downturn. We will continue utilizing internal and external research and deliberation to assess the situation, keeping our clients abreast of any change in our outlook.

This is the first time we've experienced a correction since 2011, and it has been a particularly tough quarter as a result. However, as we've discussed, we believe it's just that—a correction, and not the start of a bear market.



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“Observers are pointing to these population dynamics as an important reason why inflation and bond yields are unlikely to increase significantly in this expansionary cycle.”

Global Population Aging: Policy Implications

Aging Dynamics

Across the world, the combination of low fertility and high life expectancies is changing both population levels and age distribution. Over the last twenty-five years, countries such as Japan, Germany, Italy, and to a lesser extent the United States have reported nearly a doubling of the percent of the population over age sixty-five. This cohort could reach 25% of the total in Developed Markets (DM) by 2025. In addition, world population growth has slowed over the last thirty years, not just in countries such as Russia, which actually reported a decline, and Japan, which will soon report a decline, but in South America and Africa as well. The United States will see an aging of its population, but given births higher than deaths and net immigration the Census Bureau predicts our population will grow from 321 million to 398 million by 2050. The chart below details the more extreme

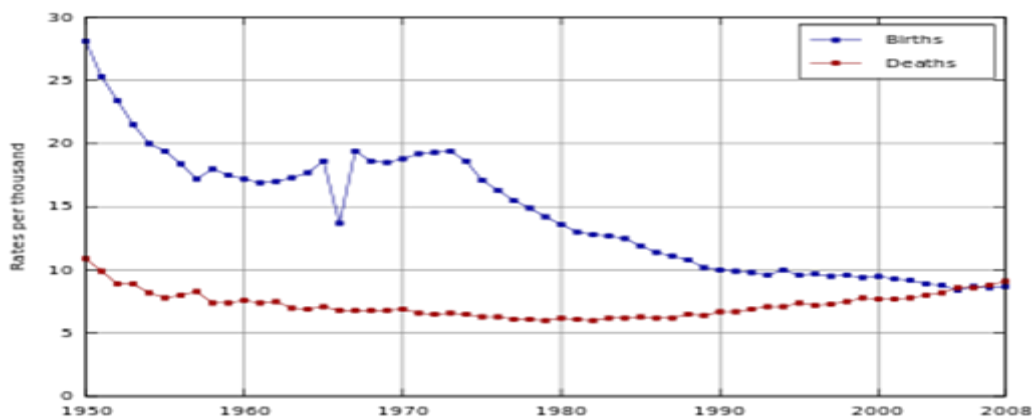
example of Japan. The impact of these demographic changes could be very significant for economic growth and interest rates: the new normal could be lower growth and interest rates through an economic cycle.

Implications

Population aging will also prompt social and economic policy adjustments. There are significant implications for public finances such as higher debt levels to fund pension and health care costs, declining labor force participation, and lower economic growth and inflation. Countries must respond to the rising tensions between two seemingly conflicting priorities --- the need to increase public spending on pension and health care and the need to reduce the growth of government spending and public sector debt. This will be a difficult political quagmire to tackle.

Public finances have been under stress due to higher debt levels associated with government deficits to ameliorate the effects of the 2007-2008 crisis. Weak economic performance has delayed the implementation of new, lower-cost health care and pension reform. The concern is that an aging population will demand more than the approximately 40% of national budgets now being devoted to these programs. In Europe, an EU Aging Work Group has made reform recommendations that are starting to be implemented, but given an aging population and resulting lower growth, these measures can only slow down the increase in debt levels, not reverse them. The charts on the next page detail expected debt and expenditure changes.

Among the nations, a differentiating factor will be the fostering of investment



Source: Wikipedia

Global Population Aging: Policy Implications

and labor productivity. This will be difficult in countries with aging populations, lower labor force participation, low productivity and net emigration. This is apparent in peripheral European countries now. Emerging Market (EM) economies could be less impacted due to both lower spending on social benefits and younger populations, but at some point social scientists expect a political consensus will emerge for higher benefits and potentially higher debt levels. The United States is not immune from the concern of an aging population and growing deficit-fueled debt increases, but population growth and productivity are

important positive considerations.

However, it is not certain these important trends will continue if immigration control is strengthened and the recent dip in productivity growth continues. Policy initiatives are crucial to foster economic growth in an aging society if we are to limit sovereign debt to manageable levels.

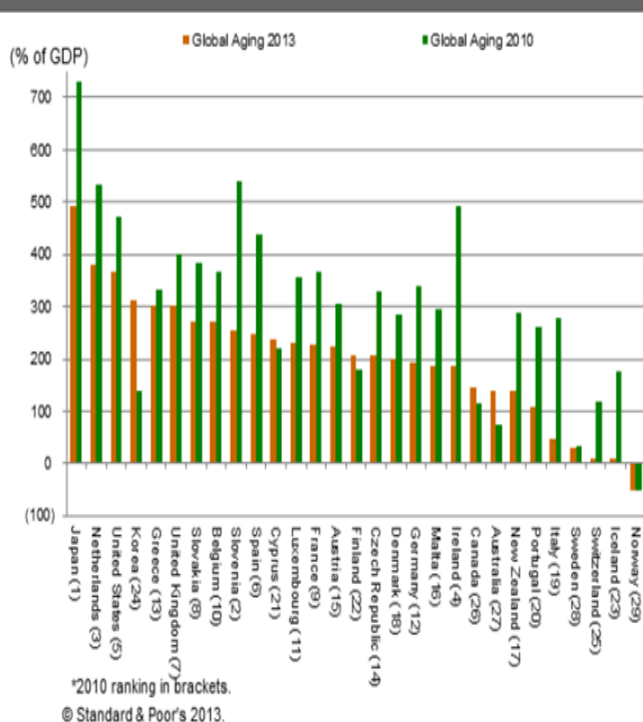
Inflation and Bond Yields

GDP growth is a function of labor force growth and productivity. In Japan we see these dynamics fostering low demand growth, including low money de-

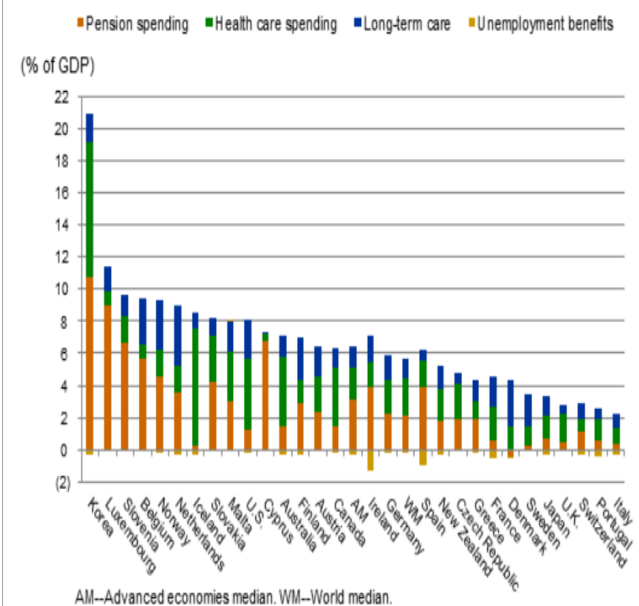
mand growth. The government has not been successful in increasing demand by its quantitative easing measures and the result is low to no inflation and the continuation of low bond yields. This outcome is also apparent, to varying degrees, in the EU and United States, where significant increases in money supply have not yet resulted in the expected growth in economic activity and inflation. We should expect increases in retirement ages, as in France and Italy, and a more concerted effort to increase productivity with targeted government education and tax policies.

Observers are pointing to these population dynamics as an important reason why inflation and bond yields are unlikely to increase significantly in this expansionary cycle. *The “new normal” could be continued low bond yields, but at least investment portfolios may not lose value as rapidly as typically feared when the Fed begins a tightening cycle.*

General Government Net Debt In 2050 In A Hypothetical No-Policy-Change Scenario--Advanced Economies*



Advanced Sovereigns Composition Of Changes In Age-Related Spending 2010-2050



“ While we believe that the liquidity issue needs to be carefully monitored, it appears that the bond market’s structure has adapted to provide sufficient liquidity...”

US Corporate Bond Market Liquidity: Should insurers be concerned?

Most corporate bonds are traded “over-the-counter” in the secondary market. Broker/Dealers may trade bonds for their own account as well as on behalf of their customers: an insurer’s investment management firm, for example. Brokers and Dealers provide liquidity to the bond market in two ways:

1. They can match a buyer and seller to complete the transaction (Agency Model) or,
2. They can buy bonds from sellers and hold them on their balance sheets until offsetting trades are found later (Principal Model)

When bonds are purchased into the Broker/Dealers account, they would bear the risk that prices fall in the interim. The former is called the “agency model” while the latter is called the

“principal model.”

What’s the Problem?

Recently, there has been much discussion arguing that bond market liquidity has been deteriorating causing concerns over liquidity. Pundits point to regulatory changes that have reduced broker/dealers’ ability and willingness to make markets, that is, hold securities for trade in their own inventory. They have shifted, rather dramatically, from the principal model toward the agency model in recent years. Financing inventory has become more expensive for dealers. Basel III capital requirements have made it more expensive for dealers to carry inventory.

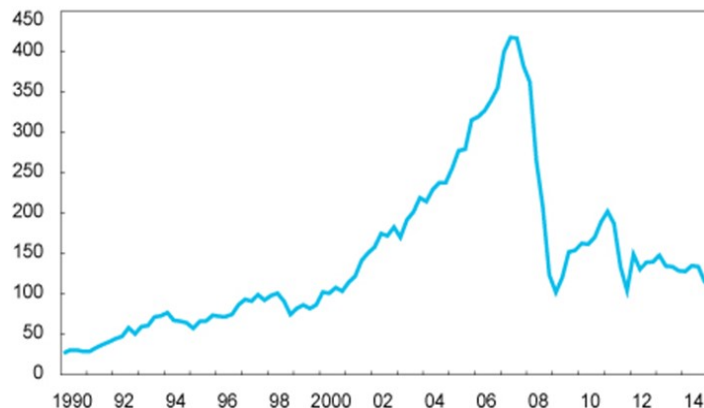
Bank market-making (Principal Model) is down significantly. Banks had provided significant liquid-

ity to the bond markets however since 2010, Dodd-Frank “Volcker rule” restrictions on bank proprietary trading have dramatically reduced this activity.

The chart below illustrates the drastic decline in inventories held by broker/dealers during the financial crisis. These inventories have been relatively stagnant since that time. Since the bond market consists of tens of thousands of outstanding corporate bond issues with varying maturity, seniority, and optionality characteristics, this diversity presents challenges for matching supply and demand. Low inventories held by broker/dealers coupled with difficulty in matching buyers and sellers drives concern that transactional spreads and timing will be adversely affected.

Dealers’ Corporate Bond Holdings Have Stagnated since the Crisis

Billions of U.S. dollars



Source: Federal Reserve Board, “Financial Accounts of the United States.”

Note: The chart shows the quantity of corporate and foreign bonds (held in the United States) owned by securities brokers and dealers.

Heightening this concern is the fact that trading volume has risen over time, especially since the financial crisis, but at a slower rate than the growth in debt outstanding. The chart below shows that U.S. corporate debt issuance has been growing rapidly due to the low interest rate environment and now fear that this regime is coming to its inevitable conclusion with the current threat of a rising interest rate environment. Corporations rushing for this cheap financing

US Corporate Bond Market Liquidity: Should insurers be concerned?

caused bond issuance to reach a record high of \$1.4 trillion in 2014. A record high of \$7.8 trillion in overall corporate debt outstanding was reached at the end of 2014.

With trading volume increasing at a slower rate, turnover rates, the ratio of trading volume to debt outstanding, for corporate bonds has declined. The current turnover rate, in the U.S. corporate bond market, is close to the levels we witnessed in 2008.

In addition to volume decreases, average trade size has been decreasing perhaps making it more difficult to execute large trades. If large trades are broken down into two smaller trades for a lower price impact, are the recorded trade volumes then inflated? If so, it would indicate a larger gap between growth in issuance and decline in actual transactional volumes. These observations are often center to the argument that bond liquidity is imperiled.

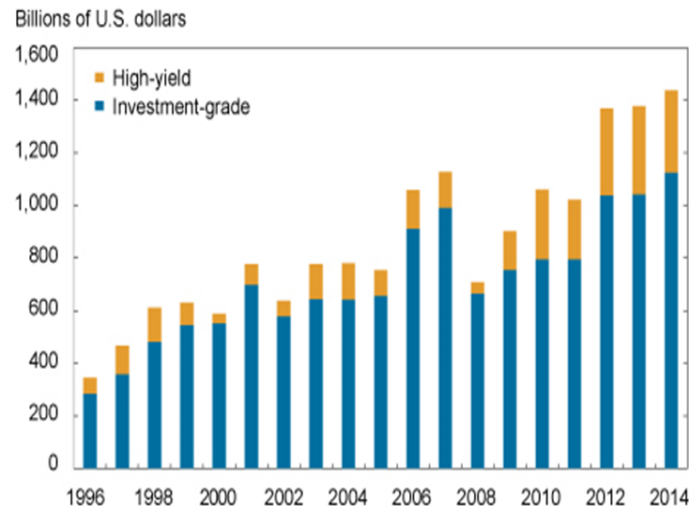
The other side of the coin

So, now let's examine the other key measurement of market liquidity: Bid/Ask Spreads. Spread represents the compensation to broker/dealers for the risk of bond price volatility over its holding period.

Since 2002, corporate bond bid-ask spreads have nar-

FIGURE 2

Corporate Bond Issuance Has Been at Record-High Levels



Source: Securities Industry and Financial Markets Association.

Note: The chart shows annual issuance of investment-grade and high-yield corporate debt.

rowed with the exception created during the financial crisis. As indicated in the

chart below, the current level of bid-ask spreads is even lower than pre-crisis levels.

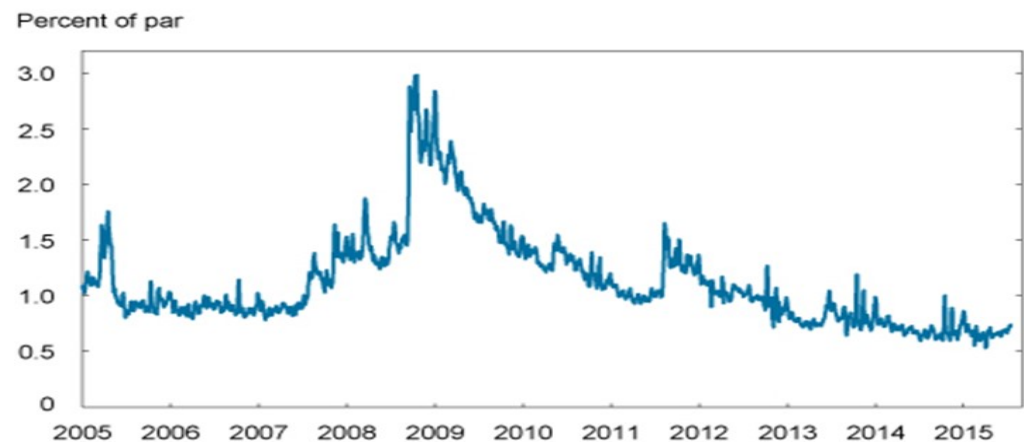
Another useful liquidity measure is the impact that a given trade has on market price. The chart on the back page, indicates a declining trade impact since the '07-08 crisis and is now well below pre-crisis levels. This along with the Bid/Ask spread compression indicates that there are no signs of a liquidity issue.

Conclusions

Conflicting signals are in play. Price-based liquidity measures like bid-ask spreads and price impact, are very low by historical standards, implying that

Figure 3¹

Bid-Ask Spreads of Corporate Bonds Have Narrowed



Source: Authors' calculations, based on Trade Reporting and Compliance Engine (TRACE) data from the Financial Industry Regulatory Authority.

Notes: The chart shows the five-day moving averages of effective bid-ask spreads. The spreads are computed daily for each investment-grade bond as the difference between the average dealer-to-client buy price and the average dealer-to-client sell price, and then averaged across bonds.

¹ 10/5/15 As published in LIBERTY STREET ECONOMICS, "Has U.S. Corporate Bond Market Liquidity Deteriorated? By Tobias Adrian, Michael Fleming, Or Shachar, and Erik Vogt

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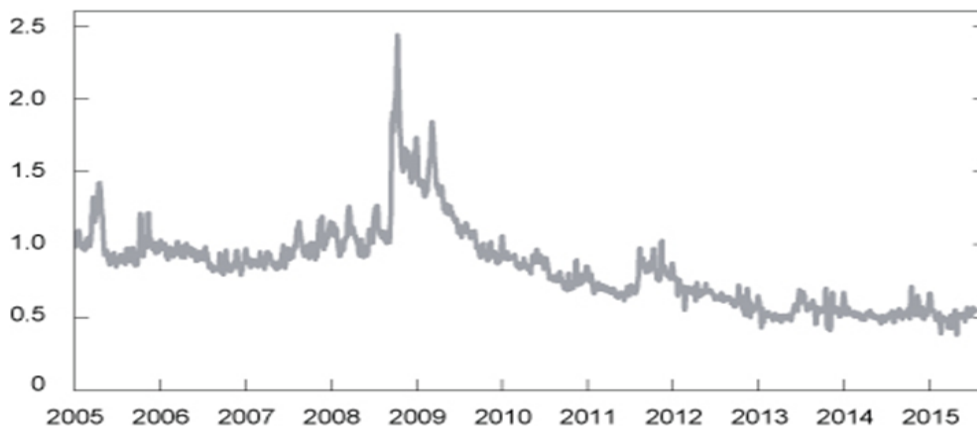
US Corporate Bond Market Liquidity: Should insurers be concerned?

there is no liquidity issue. This is surprising as it offsets concerns driven by declines in broker/dealer ownership of corporate bonds (Principal Model) and possible reductions in trade size. While we believe that the liquidity issue needs to be carefully monitored, it appears that the bond market's structure has adapted to provide sufficient liquidity at pre-

sent, perhaps through additional market participants such as hedge funds and high-frequency-trading firms. We consider the question of whether liquidity in US corporate bond market is of concern to be unresolved at this time and call for additional research and vigilance on the matter.

Price Impact Has Declined Markedly

Percent of par per 100 million U.S. dollars



Source: Authors' calculations, based on Trade Reporting and Compliance Engine (TRACE) data from the Financial Industry Regulatory Authority.

Notes: The chart shows the five-day moving average of price impact. Price impact is calculated daily for each investment-grade bond as the absolute price return divided by dollar volume, and then averaged across securities.



Upcoming Events

The Cayman Captive Forum takes place on Dec 1-3 at the Ritz Carlton in Grand Cayman. Carl Terzer will be in attendance and hopes to see you there!



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